Lab sheet-8

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```
import os

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.metrics import (

    accuracy_score,

    precision_score,

    recall_score,

    f1_score,

    confusion_matrix,

    classification_report

)

# Step 0: Setup Output Directory

output_dir = "churn_output_images"

os.makedirs(output_dir, exist_ok=True)

# Step 1 & 2: Load Dataset from GitHub

DATA_URL = "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-
icp4d/master/data/Telco-Customer-Churn.csv"

try:

    df = pd.read_csv(DATA_URL)

    print("Dataset loaded successfully.")

except Exception as err:

    print(f"Failed to fetch dataset: {err}")

    raise SystemExit

print(f"Dataset Dimensions: {df.shape}")
```

```
print(df.head(3))
# Step 3: Data Preprocessing & Feature Encoding
# Drop customer identifier
df = df.drop(columns=['customerID'])
# Convert TotalCharges from string to numeric (handles blank spaces as NaN)
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'].str.strip(), errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(df['TotalCharges'].median())
# Map binary target variable
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})
# Identify binary columns vs. multi-class categorical columns
binary_cols = ['Partner', 'Dependents', 'PhoneService', 'PaperlessBilling']
for col in binary_cols:
    df[col] = df[col].map({'Yes': 1, 'No': 0})
# Map gender
df['gender'] = df['gender'].map({'Male': 1, 'Female': 0})
# One-Hot Encode remaining categorical variables
categorical_cols = [
    'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup',
    'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies',
    'Contract', 'PaymentMethod'
]
df = pd.get_dummies(df, columns=categorical_cols, drop_first=True)
# Separate features (X) and target (y)
X = df.drop(columns=['Churn'])
y = df['Churn']
# Step 4: Train-Test Split (80:20 Ratio)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
```

```
)
print(f"Training Samples: {X_train.shape[0]} | Testing Samples: {X_test.shape[0]}")
# Step 5: Train Decision Tree Classifier
# Using max_depth=4 to balance expressiveness with interpretability and avoid overfitting
dt_classifier = DecisionTreeClassifier(
    criterion='gini',
    max_depth=4,
    min_samples_split=20,
    min_samples_leaf=10,
    random_state=42
)
dt_classifier.fit(X_train, y_train)
# Step 6 & 7: Model Predictions & Performance Evaluation
y_pred = dt_classifier.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)
print("\n" + "=" * 55)
print("MODEL PERFORMANCE METRICS")
print("=" * 55)
print(f"Accuracy  : {accuracy:.4f}")
print(f"Precision : {precision:.4f}")
print(f"Recall    : {recall:.4f}")
print(f"F1-Score  : {f1:.4f}")
print("\nDetailed Classification Report:")
print(classification_report(y_test, y_pred, target_names=['Retained (0)', 'Churned (1)']))
```

```
# Plot and Save Confusion Matrix
plt.figure(figsize=(6, 5))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['Retained', 'Churned'],
    yticklabels=['Retained', 'Churned']
)
plt.title('Confusion Matrix - Customer Churn', fontsize=12, fontweight='bold')
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')
plt.tight_layout()
cm_path = os.path.join(output_dir, "confusion_matrix.png")
plt.savefig(cm_path, dpi=300, bbox_inches='tight')
print(f"[Saved Image] Confusion matrix saved to: {cm_path}")
plt.show()
# Step 8: Visualize the Decision Tree Structure
plt.figure(figsize=(24, 12))
plot_tree(
    dt_classifier,
    feature_names=X.columns.tolist(),
    class_names=['Retained', 'Churned'],
    filled=True,
    rounded=True,
    fontsize=9
)
```

```
plt.title('Trained Decision Tree (max_depth=4)', fontsize=15, fontweight='bold')
plt.tight_layout()
tree_path = os.path.join(output_dir, "decision_tree_structure.png")
plt.savefig(tree_path, dpi=300, bbox_inches='tight')
print(f"[Saved Image] Decision tree visualization saved to: {tree_path}")
plt.show()
# Step 9: Feature Importance Analysis
importance_df = pd.DataFrame({
    'Feature': X.columns,
    'Importance': dt_classifier.feature_importances_
}).sort_values(by='Importance', ascending=False)
# Keep only top 8 features contributing to splits
top_features = importance_df[importance_df['Importance'] > 0.01]
plt.figure(figsize=(8, 4.5))
sns.barplot(
    data=top_features,
    x='Importance',
    y='Feature',
    palette='viridis'
)
plt.title('Key Features Driving Customer Churn', fontsize=12, fontweight='bold')
plt.xlabel('Feature Importance Score')
plt.ylabel('Attribute')
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
fi_path = os.path.join(output_dir, "feature_importance.png")
plt.savefig(fi_path, dpi=300, bbox_inches='tight')
print(f"[Saved Image] Feature importances saved to: {fi_path}")
```

plt.show()

Step 10: Print Strategic Business Recommendations

print("\n" + "=" * 55)

print("BUSINESS RETENTION RECOMMENDATIONS")

print("=" * 55)

print("1. Target Month-to-Month Contracts: Contract type is the strongest split. Transition customers toward annual contracts with incentives.")

print("2. Monitor Tenure Milestones: Churn risk drops sharply as customer tenure passes 12 months; provide dedicated onboarding support in months 1-6.")

print("3. TechSupport & Fiber Quality: Customers on Fiber Optic without TechSupport exhibit high churn; bundle assistance and resolve network latency.")

```
Dataset loaded successfully.
Dataset Dimensions: (7043, 21)
   customerID  gender  SeniorCitizen  ... MonthlyCharges TotalCharges  Churn
0  7590-VHVEG  Female              0  ...          29.85        29.85     No
1  5575-GNVDE    Male              0  ...          56.95       1889.5     No
2  3668-QPYBK    Male              0  ...          53.85       108.15    Yes

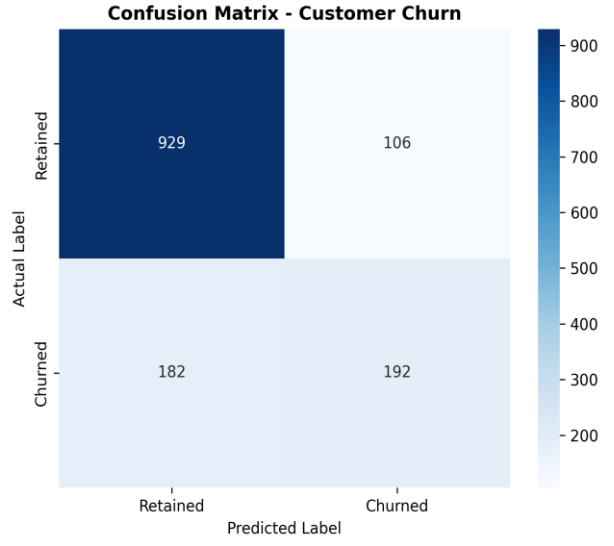
[3 rows x 21 columns]
Training Samples: 5634 | Testing Samples: 1409

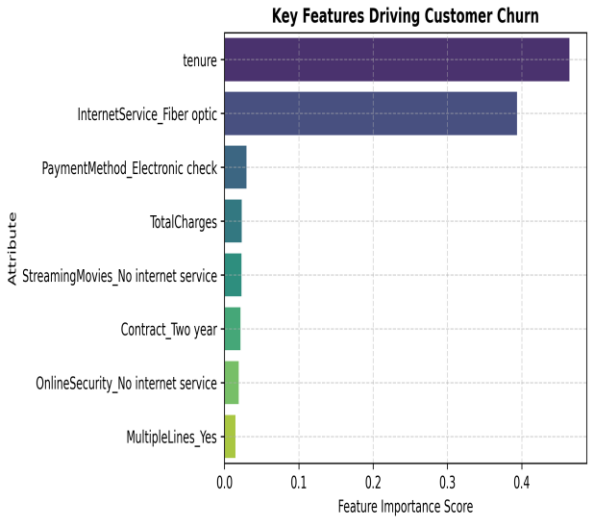
==================================================
MODEL PERFORMANCE METRICS
==================================================
Accuracy  : 0.7956
Precision : 0.6443
Recall    : 0.5134
F1-Score  : 0.5714

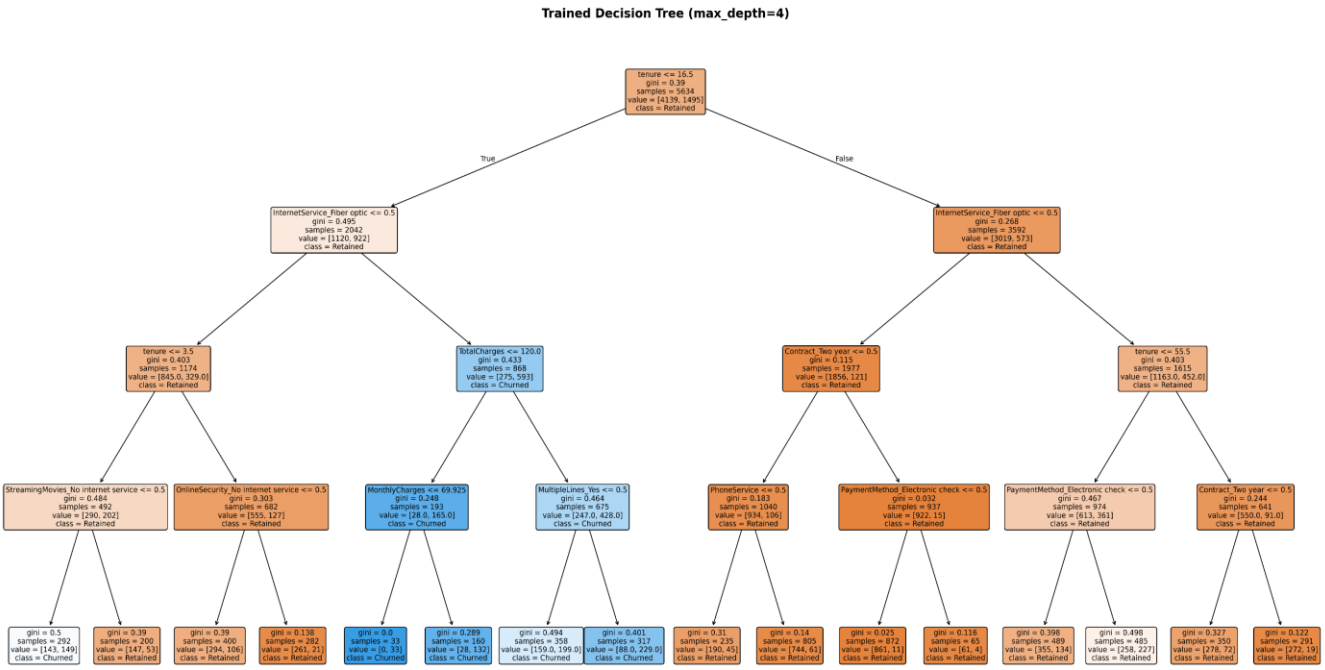
Detailed Classification Report:
              precision    recall  f1-score   support

Retained (0)       0.84      0.90      0.87      1035
 Churned (1)       0.64      0.51      0.57       374

    accuracy                           0.80      1409
   macro avg       0.74      0.71      0.72      1409
weighted avg       0.79      0.80      0.79      1409
```







Experiment Questions

1. What is classification, and how does it differ from regression?

- **Classification:** A supervised learning task where the target output variable y is discrete and categorical (e.g., predicting whether a customer will churn or stay: $y \in \{0,1\}$).
- **Regression:** A supervised learning task where the target output variable y is continuous and numeric (e.g., predicting customer lifetime value, tenure duration in days, or revenue).

2. Explain the working principle of the Decision Tree algorithm.

The Decision Tree builds a decision model top-down through recursive binary splitting:

1. It scans all candidate features and evaluates every possible numerical or categorical split threshold.
2. It chooses the split that yields the largest reduction in node impurity (maximizing Information Gain or Gini reduction).
3. The dataset is partitioned into subsets at internal decision nodes.
4. This splitting repeats recursively until a stopping condition is reached (e.g., reaching max_depth, node purity, or a minimum sample threshold).
5. Leaf nodes assign predictions based on the majority class among the training points landing in that terminal region.

3. Differentiate between Gini Index and Entropy as splitting criteria.

Attribute	Gini Index (Impurity) PDF	Entropy PDF
Formula	$1 - \sum_{i=1}^{C} p_i^2$	$-\sum_{i=1}^{C} p_i \log_2(p_i)$
Computation Speed	Faster, as it involves simple arithmetic squaring.	Slower, as it requires computing logarithms for every candidate split.
Output Range	Bounds between 0.0 (pure) and 0.5 (equal binary split).	Bounds between 0.0 (pure) and 1.0 (equal binary split).
Sensitivity	Slightly more sensitive to isolating the largest class into its own node.	Tends to generate slightly more balanced tree structures.

4. What is a Confusion Matrix? Explain its components.

A Confusion Matrix is a tabular layout that compares a classification model's predicted labels against ground-truth labels across four quadrants:

- **True Positive (TP):** The model correctly predicted a churning customer as Churned.
- **True Negative (TN):** The model correctly predicted a retained customer as Retained.
- **False Positive (FP / Type I Error):** The model predicted Churned, but the customer actually stayed.
- **False Negative (FN / Type II Error):** The model predicted Retained, but the customer actually churned.

5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?

- **Accuracy:** $\frac{TP+TN}{TP+TN+FP+FN}$ — Ratio of total correct predictions to all samples.
- **Precision:** $\frac{TP}{TP+FP}$ — The proportion of predicted churners that actually churned.
- **Recall (Sensitivity):** $\frac{TP}{TP+FN}$ — The proportion of actual churners correctly detected by the model.
- **F1-Score:** $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$ — Harmonic mean of Precision and Recall.

- **Importance:** In churn detection, datasets are class-imbalanced. Relying only on accuracy allows a naive model that predicts "No Churn" for everyone to score high accuracy while missing every churner; tracking Recall and F1-Score ensures actual churn events are detected.

6. What is overfitting in a Decision Tree? How can it be reduced?

- **Overfitting:** Occurs when a decision tree grows without constraints, splitting until it memorizes noise and sample-specific idiosyncrasies of the training data, leading to poor generalization on unseen test data.
- **How to Reduce It:**
    - **Pre-pruning (Hyperparameter Constraints):** Set limits on max_depth, require a minimum number of samples to split (min_samples_split), or enforce minimum samples per leaf (min_samples_leaf).
    - **Post-pruning (Cost-Complexity Pruning):** Grow a full tree and prune sub-trees back using a cost-complexity parameter (α).
    - **Ensemble Methods:** Use bagging or boosting ensembles like Random Forests or Gradient Boosted Trees.

7. Why is customer churn prediction important for businesses?

Acquiring new customers involves high marketing and sales acquisition costs (CAC) compared to maintaining an existing account. Early churn prediction provides businesses with an operational window to proactively resolve customer dissatisfaction, offer customized discounts, and preserve recurring revenue streams before the contract is officially terminated.

8. What are the advantages and limitations of the Decision Tree algorithm?

- **Advantages:**
    - High interpretability: Rules can be visualized and explained directly to business stakeholders.
    - Requires minimal data preparation: Inherently handles both numerical and categorical variables without mandatory feature normalization.
    - Captures non-linear interactions between attributes naturally.
- **Limitations:**
    - High variance: Small changes in training data can generate an entirely different tree structure.
    - Susceptible to overfitting when unpruned.

- Orthogonal decision boundaries: Trees partition feature space strictly parallel to coordinate axes, struggling with diagonal decision boundaries.

9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.

- **Credit Risk Assessment & Loan Default Approval:** Classifying whether loan applicants pose low or high credit default risk based on credit scores, income, and debt ratios.
- **Medical Diagnosis:** Assisting doctors in classifying whether a tumor is benign or malignant, or detecting diabetes risk from metabolic indicators.
- **Fraud Detection:** Identifying fraudulent credit card transactions based on transaction frequency, amount thresholds, and geographic discrepancies.

10. How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?

- **Targeted Interventions:** Identifies high-risk customer cohorts (e.g., month-to-month contracts with fiber optic service), allowing retention teams to deploy proactive concessions instead of blanket discounts.
- **Resource Optimization:** Focuses customer service spend and retention incentives only on at-risk accounts with high lifetime value rather than wasting budget on satisfied customers.
- **Product & Service Remediation:** Highlights underlying operational failures (e.g., if lacking tech support consistently drives churn, leadership can improve technical support staffing and onboarding guides).

Experiment No. 8: Customer Churn Prediction using Decision Tree Classification

Aim

To build and evaluate a Decision Tree Classification model for predicting customer churn using a real-world dataset and analyze the factors influencing customer retention.

Questions & Answers

Q1. What is classification, and how does it differ from regression?

Answer:

- **Classification:** A supervised learning task that maps input features to categorical class labels (e.g., *Churn* vs. *No Churn*).
- **Regression:** A supervised learning task that maps input features to a continuous numerical target (e.g., *Predicting House Prices*).

Aspect	Classification	Regression
Output Type	Discrete / Categorical labels	Continuous numerical values
Example	Email Spam vs. Not Spam	Predicting future stock price
Primary Evaluation Metrics	Accuracy, Precision, Recall, F1-Score	MAE, MSE, RMSE, R^2 Score

Q2. Explain the working principle of the Decision Tree algorithm.

Answer:

A Decision Tree builds a flowchart-like tree structure by recursively partitioning the training dataset into smaller subsets:

- **Root Node Selection:** Evaluates candidate feature split points using impurity criteria (Gini Index or Entropy) to select the optimal root node feature.
- **Splitting Phase:** Splits data into branches based on feature threshold conditions.
- **Recursive Partitioning:** Repeats the splitting process on child sub-nodes.
- **Terminal / Leaf Nodes:** Stops splitting upon hitting predefined stopping criteria (e.g., max_depth, min_samples_split) and assigns class labels via majority voting.

Q3. Differentiate between Gini Index and Entropy as splitting criteria.

Answer:

Attribute	Gini Index	Entropy
Formula	1 - \sum_{i=1}^{C} (p_i)^2	-\sum_{i=1}^{C} p_i \log_2(p_i)
Range	0 to 0.5 (for binary classification)	0 to 1 (for binary classification)
Computational Speed	Faster (no logarithmic calculations required)	Slower (requires log calculations)
Default Usage	Default impurity measure in Scikit-learn	Preferred when measuring Information Gain explicitly

Q4. What is a Confusion Matrix? Explain its components.

Answer:

A **Confusion Matrix** is a 2 \times 2 performance evaluation table comparing actual target classes with predicted classes:

- **True Positive (TP):** Model correctly predicted a churning customer as *Churn*.
- **True Negative (TN):** Model correctly predicted a retained customer as *No Churn*.
- **False Positive (FP - Type I Error):** Model incorrectly flagged a retained customer as *Churn*.
- **False Negative (FN - Type II Error):** Model incorrectly flagged an actual churning customer as *No Churn*.

Q5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?

Answer:

- **Accuracy:** Overall proportion of correct predictions across all classes.

\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}

- **Precision:** Ratio of correctly predicted positive observations to total predicted positives.

\text{Precision} = \frac{TP}{TP + FP}

- **Recall (Sensitivity):** Ratio of correctly predicted positive observations to all actual positive instances.

\text{Recall} = \frac{TP}{TP + FN}

- **F1-Score:** Harmonic mean of Precision and Recall, crucial for handling imbalanced datasets.

\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}

Q6. What is overfitting in a Decision Tree? How can it be reduced?

Answer:

- **Overfitting:** Occurs when a Decision Tree grows unconstrained, memorizing noisy training data details and creating overly complex decision boundaries. This results in high training accuracy but poor generalization on test data.
- **Reduction Strategies:**
- **Pre-Pruning:** Setting hyperparameter limits like max_depth, min_samples_split, or min_samples_leaf.
- **Post-Pruning:** Trimming branches after full tree growth using Cost Complexity Pruning (ccp_alpha).
- **Ensemble Methods:** Using Random Forests or Gradient Boosted Trees.

Q7. Why is customer churn prediction important for businesses?

Answer:

- **Cost Efficiency:** Acquiring a new customer is up to 5x to 25x more expensive than retaining an existing customer.
- **Revenue Protection:** Prevents recurring revenue losses from high-margin subscription accounts.
- **Proactive Interventions:** Allows marketing teams to issue targeted retention offers before customers formally leave.

Q8. What are the advantages and limitations of the Decision Tree algorithm?

Answer:

- **Advantages:**
- High interpretability; decision rules can be easily visualized and explained.

- Requires minimal data preprocessing (handles numerical and categorical features naturally).
- Non-parametric; makes no strict assumptions about underlying data distributions.
- **Limitations:**
- Highly prone to overfitting if not properly pruned.
- Unstable; small changes in input data can yield vastly different tree structures.
- Can struggle with complex non-linear relationships compared to gradient boosting.

Q9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.

Answer:

- **Banking & Credit Scoring:** Evaluating loan applicants as *High Risk* or *Low Risk*.
- **Healthcare & Diagnosis:** Classifying medical test results to detect specific diseases.
- **Cybersecurity:** Classifying incoming network traffic as *Legitimate* or *Malicious Intrusion*.

Q10. How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?

Answer:

- **Targeted Incentives:** Identifies high-churn risk segments to send customized discount vouchers or loyalty extensions rather than blanket promotions.
- **Product Enhancements:** Reveals key churn drivers (e.g., month-to-month contracts or lack of tech support), allowing business teams to optimize product tiers and improve user onboarding.
- **Maximizing Customer Lifetime Value (CLV):** Extends average account lifespan, driving sustainable profit margins.